

# Best Programming Practice

1. All values as variables including Fixed, User Inputs, and Results
2. Proper naming conventions for all variables

```
String name = "Eric";  
double height = input.nextDouble();  
double totalDistance = distanceFromToVia + distanceViaToFinalCity;
```

3. Proper Program Name and Class Name
4. Follow proper indentation
5. Give comments for every step or logical block like a variable declaration or conditional and loop blocks

1. **Sample Program 1** - Create a program to check if 3 values are internal angles of a triangle.

IMP => Follow Good Programming Practice demonstrated below in all Practice Programs

**Hint =>**

- a. Get integer input for 3 variables named x, y, and z.
- b. Find the sum of x, y, and z.
- c. If the sum is equal to 180, print "The given angles are internal angles of a triangle" else print They are not

Java

```
// Creating Class with name TriangleChecker indicating the purpose is to  
// check if the internal angles add to 180  
import java.util.Scanner;  
  
class TriangleChecker {  
    public static void main(String[] args) {  
        // Create a Scanner Object  
        Scanner input = new Scanner(System.in);  
  
        // Get 3 input values for angles  
        int x = input.nextInt();  
        int y = input.nextInt();  
        int z = input.nextInt();  
  
        // Find the sum of all angles  
        int sumOfAngles = x + y + z;  
  
        // Check if sum is equal to 180 and print either true or false  
        System.out.println("The given angles " +x+ ", " +y+ ", " + z +  
            " add to " + sumOfAngles);
```

```

if (sumOfAngles == 180) {
    System.out.println("The given angles are internal angles of a " +
        "Triangle");
} else {
    System.out.println("The given angles are not internal angles " +
        "of a Triangle");
}

// Closing the Scanner Stream
input.close();
}
}

```

2. **Sample Program 2** - Create a program to find the sum of all the digits of a number given by a user.

**Hint =>**

- a. Get an integer input for the number variable.
- b. Create an integer variable sum with an initial value of 0.
- c. Create a while loop to access each digit of the number.
- d. Inside the loop, add each digit of the number to the sum.
- e. Finally, print the sum outside the loop

Java

```

// Create SumOfDigit Class to compute the sum of all digits of a number
import java.util.Scanner;

class SumOfDigits {

    public static void main(String[] args) {
        // Create a Scanner Object
        Scanner input = new Scanner(System.in);

        // Get input value for number
        int origNumber = input.nextInt();

        // Define variable number and sum initialized to zero
        int number = origNumber;
        int sum = 0;
    }
}

```

```
// Run while loop to access each digit of number
while (number != 0) {
    // Use number % 10 to find each digit of number from last
    int digit = number % 10;

    // add each digit to sum
    sum += digit;

    // Remove last digit from number essentially get the quotient
    number = number / 10;
}

// Print the sum and close the Scanner Stream
System.out.println("The sum of digit of number:" +origNumber+ " = " +
                    sum);
input.close();
}
}
```

## Level 2 Practice Programs

1. Write a LeapYear program that takes a year as input and outputs the Year is a Leap Year or not a Leap Year.

**Hint =>**

- a. The LeapYear program only works for year  $\geq 1582$ , corresponding to a year in the Gregorian calendar. So ensure to check for the same.
- b. Further, the Leap Year is a Year divisible by 4 and not 100 unless it is divisible by 400. E.g. 1800 is not a Leap Year and 2000 is a Leap Year.
- c. Write code having multiple **if else** statements based on conditions provided above and a second part having only one if statement and multiple logical

```
import java.util.Scanner;
public class LeapYear {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Take user input for the year
        System.out.print("Enter a year (>= 1582): ");
        int year = input.nextInt();

        // Check if the year is valid (Gregorian calendar starts from 1582)
        if (year >= 1582) {

            // Part 1: Using multiple if-else statements
            boolean isLeapYear = false;
            if (year % 4 == 0) {
                if (year % 100 == 0) {
                    if (year % 400 == 0) {
                        isLeapYear = true;
                    }
                    else {
                        isLeapYear = false;
                    }
                }
                else {
                    isLeapYear = true;
                }
            }
        }
    }
}
```

```

        else {
            isLeapYear = false;
        }

        // Output the result using multiple if-else statements
        if (isLeapYear) {
            System.out.println("The year " + year + " is a Leap Year
(using if-else).");
        }
        else {
            System.out.println("The year " + year + " is not a Leap
Year (using if-else).");
        }

        // Part 2: Using a single if statement with multiple logical
conditions
        if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
            System.out.println("The year " + year + " is a Leap Year
(using single if).");
        }
        else {
            System.out.println("The year " + year + " is not a Leap
Year (using single if).");
        }
    }
    else {
        System.out.println("The year " + year + " is not valid. Please
enter a year >= 1582.");
    }
    // Close the scanner
    input.close();
}
}

```

2. Rewrite program 1 to determine Leap Year with single if condition using logical and **&&** and or **||** operators

```

import java.util.Scanner;
public class LeapYear2 {

```

```
public static void main(String[] args) {
    // Create a Scanner object to take user input
    Scanner input = new Scanner(System.in);
    // Take user input for the year
    System.out.print("Enter a year (>= 1582): ");
    int year = input.nextInt();
    // Check if the year is valid (Gregorian calendar starts from
1582)

    if (year >= 1582) {
        // Determine if the year is a Leap Year using a single if
condition with logical operators
        if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
            System.out.println("The year " + year + " is a Leap Year.");
        }
        else {
            System.out.println("The year " + year + " is not a Leap
Year.");
        }
    }
    else {
        System.out.println("The year " + year + " is not valid.
Please enter a year >= 1582.");
    }
    // Close the scanner
    input.close();
}
}
```

3. Write a program to input marks and 3 subjects physics, chemistry and maths. Compute the percentage and then calculate the grade as per the following guidelines

| Grade | Remarks                                                       | Marks         |
|-------|---------------------------------------------------------------|---------------|
| A     | (Level 4, above agency-normalized standards)                  | 80% and above |
| B     | (Level 3, at agency-normalized standards)                     | 70-79%        |
| C     | (Level 2, below, but approaching agency-normalized standards) | 60-69%        |
| D     | (Level 1, well below agency-normalized standards)             | 50-59%        |
| E     | (Level 1- , too below agency-normalized standards)            | 40-49%        |
| R     | (Remedial standards)                                          | 39% and below |

Hint =>

- Ensure the Output clearly shows the Average Mark as well as the Grade and Remarks

```
import java.util.Scanner;
public class StudentPerformance {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Take user input for marks in three subjects
        System.out.print("Enter marks for Physics: ");
        int physics = input.nextInt();
        System.out.print("Enter marks for Chemistry: ");
        int chemistry = input.nextInt();
        System.out.print("Enter marks for Maths: ");
        int maths = input.nextInt();

        // Calculate the total and percentage
        int total = physics + chemistry + maths;
        double percentage = total / 3.0;
        // Determine the grade and remarks
        char grade;
        String remarks;
        if (percentage >= 80) {
            grade = 'A';
            remarks = "Level 4, above agency-normalized standards";
        }
        else if (percentage >= 70) {
            grade = 'B';
```

```

        remarks = "Level 3, at agency-normalized standards";
    }
    else if (percentage >= 60) {
        grade = 'C';
        remarks = "Level 2, below, but approaching
agency-normalized standards";
    }
    else if (percentage >= 50) {
        grade = 'D';
        remarks = "Level 1, well below agency-normalized
standards";
    }
    else if (percentage >= 40) {
        grade = 'E';
        remarks = "Level 1-, too below agency-normalized
standards";
    }
    else {
        grade = 'R';
        remarks = "Remedial standards";
    }
}

// Print the results
System.out.println("Average Mark: " + percentage);
System.out.println("Grade: " + grade);
System.out.println("Remarks: " + remarks);
// Close the scanner
input.close();
}
}

```

4. Write a Program to check if the given number is a prime number or not

**Hint =>**

- A number that can be divided exactly only by itself and 1 are Prime Numbers,
- Prime Numbers checks are done for numbers greater than 1
- Loop through all the numbers from 2 to the user input number and check if the remainder is zero. If the remainder is zero break out from the loop as the number is divisible by some other number and is not a prime number.
- Use isPrime boolean variable to store the result



```
import java.util.Scanner;
public class PrimeNumberCheck {

    public static void main(String[] args) {
        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);
        // Take user input for the number
        System.out.print("Enter a number: ");
        int number = input.nextInt();
        // Initialize the isPrime variable
        boolean isPrime = true;
        // Check if the number is greater than 1
        if (number <= 1) {
            isPrime = false;
        }
        else {
            // Loop through all the numbers from 2 to the user input
            number
            for (int i = 2; i < number; i++) {
                // Check if the remainder is zero
                if (number % i == 0) {
                    // The number is divisible by some other
                    number, so it's not a prime number
                    isPrime = false;
                    break;
                }
            }
        }
        // Print if the number is prime
        if (isPrime) {
            System.out.println("The number " + number + " is a prime
            number.");
        }
        else {
            System.out.println("The number " + number + " is not a
            prime number.");
        }
        // Close the scanner
        input.close();
    }
}
```

5. Write a program FizzBuzz, take a number as user input, and if it is a positive integer loop from 0 to the number and print the number, but for multiples of 3 print "Fizz" instead of the number, for multiples of 5 print "Buzz", and for multiples of both print "FizzBuzz".

**Hint =>**

- a. Write the program and use **for** loop

```
import java.util.Scanner;
public class FizzBuzz {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Take user input for the number
        System.out.print("Enter a positive integer: ");
        int number = input.nextInt();

        // Check if the user has entered a positive integer
        if (number > 0) {
            // Loop from 0 to the entered number
            for (int i = 0; i <= number; i++) {
                // Check for multiples of 3 and 5
                if (i % 3 == 0 && i % 5 == 0) {
                    System.out.println("FizzBuzz");
                }
                else if (i % 3 == 0) {
                    System.out.println("Fizz");
                }
                else if (i % 5 == 0) {
                    System.out.println("Buzz");
                }
                else {
                    System.out.println(i);
                }
            }
        }
        else {
```

```

        System.out.println("Please enter a positive integer.");
    }

    // Close the scanner
    input.close();
}
}

```

## 6. Rewrite the program 5 FizzBuzz using while loop

```

import java.util.Scanner;
public class FizzBuzz2 {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Take user input for the number
        System.out.print("Enter a positive integer: ");
        int number = input.nextInt();

        // Check if the user has entered a positive integer
        if (number > 0) {
            int i = 0; // Initialize the counter

            // Loop from 0 to the entered number using while loop
            while (i <= number) {

                // Check for multiples of 3 and 5
                if (i % 3 == 0 && i % 5 == 0) {
                    System.out.println("FizzBuzz");
                }
                else if (i % 3 == 0) {
                    System.out.println("Fizz");
                }
                else if (i % 5 == 0) {
                    System.out.println("Buzz");
                }
            }
        }
    }
}

```

```

        else {
            System.out.println(i);
        }

        // Increment the counter
        i++;
    }
}
else {
    System.out.println("Please enter a positive integer.");
}

// Close the scanner
input.close();
}
}

```

## 7. Create a program to find the BMI of a person

### Hint =>

- Take user input in double for the weight (in kg) of the person and height (in cm) for the person and store it in the corresponding variable.
- Use the formula  $BMI = \text{weight} / (\text{height} * \text{height})$ . Note unit is  $\text{kg}/\text{m}^2$ . For this convert cm to meter
- Use the table to determine the weight status of the person

| BMI         | Status      |
|-------------|-------------|
| $\leq 18.4$ | Underweight |
| 18.5 - 24.9 | Normal      |
| 25.0 - 39.9 | Overweight  |
| $\geq 40.0$ | Obese       |

```
import java.util.Scanner;
public class BMICalculator {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Take user input for weight in kg
        System.out.print("Enter weight in kg: ");
        double weight = input.nextDouble();

        // Take user input for height in cm
        System.out.print("Enter height in cm: ");
        double heightCm = input.nextDouble();

        // Convert height from cm to meters
        double heightM = heightCm / 100;

        // Calculate BMI
        double bmi = weight / (heightM * heightM);

        // Check weight status based on BMI
        String status;
        if (bmi <= 18.4) {
            status = "Underweight";
        }
        else if (bmi <= 24.9) {
            status = "Normal";
        }
        else if (bmi <= 39.9) {
            status = "Overweight";
        }
        else {
            status = "Obese";
        }
        // Print the BMI and weight status
        System.out.println("BMI: " + bmi);
        System.out.println("Status: " + status);
        // Close the scanner
        input.close();
    }
}
```

```
}
```

8. Create a program to find the youngest friends among 3 Amar, Akbar, and Anthony based on their ages and the tallest among the friends based on their heights

**Hint =>**

- Take user input for the age and height of the 3 friends and store it in a variable
- Find the smallest of the 3 ages to find the youngest friend and display it
- Find the largest of the 3 heights to find the tallest friend and display it

```
import java.util.Scanner;
public class HeightComparison {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Take user input for ages
        System.out.print("Enter the age of Amar: ");
        int ageAmar = input.nextInt();
        System.out.print("Enter the age of Akbar: ");
        int ageAkbar = input.nextInt();
        System.out.print("Enter the age of Anthony: ");
        int ageAnthony = input.nextInt();

        // Take user input for heights
        System.out.print("Enter the height of Amar (in cm): ");
        double heightAmar = input.nextDouble();
        System.out.print("Enter the height of Akbar (in cm): ");
        double heightAkbar = input.nextDouble();
        System.out.print("Enter the height of Anthony (in cm): ");
        double heightAnthony = input.nextDouble();

        // Finding the youngest friend
        String youngestFriend;
        int youngestAge = Math.min(ageAmar, Math.min(ageAkbar,
ageAnthony));
        if (youngestAge == ageAmar) {
            youngestFriend = "Amar";
        }
    }
}
```

```

        else if (youngestAge == ageAkbar) {
            youngestFriend = "Akbar";
        }
        else {
            youngestFriend = "Anthony";
        }

        // Finding the tallest friend
        String tallestFriend;
        double tallestHeight = Math.max(heightAmar,
Math.max(heightAkbar, heightAnthony));
        if (tallestHeight == heightAmar) {
            tallestFriend = "Amar";
        }
        else if (tallestHeight == heightAkbar) {
            tallestFriend = "Akbar";
        }
        else {
            tallestFriend = "Anthony";
        }
        // Print the results

        System.out.println("The youngest friend is " + youngestFriend +
" with age " + youngestAge);
        System.out.println("The tallest friend is " + tallestFriend + "
with height " + tallestHeight + " cm");
        // Close the scanner
        input.close();
    }
}

```

9. Create a program to print the greatest factor of a number beside itself using a loop.

**Hint =>**

- Get an integer input and assign it to the number variable. As well as define a greatestFactor variable and assign it to 1
- Create a **for** loop that runs from last but one till 1 as in  $i = \text{number} - 1$  to  $i = 1$ .
- Inside the loop, check if the number is perfectly divisible by  $i$  then assign  $i$  to greatestFactor variable and break the loop.
- Display the greatestFactor variable outside the loop

```
import java.util.Scanner;
public class GreatestFactor {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Take user input for the number
        System.out.print("Enter an integer: ");
        int number = input.nextInt();

        // Initialize the greatestFactor variable
        int greatestFactor = 1;

        // Loop from the last but one till 1
        for (int i = number - 1; i >= 1; i--) {

            // Check if the number is perfectly divisible by i
            if (number % i == 0) {
                greatestFactor = i;
                break;
            }
        }

        // Print the greatest factor
        System.out.println("The greatest factor of " + number + "
besides itself is " + greatestFactor);

        // Close the scanner
        input.close();
    }
}
```

10. Create a program to find the power of a number.

**Hint =>**

- Get integer input for two variables named number and power.
- Create a result variable with an initial value of 1.
- Run a for loop from i = 1 to i <= power.
- In each iteration of the loop, multiply the result with the number and assign the value to the result.
- Finally, print the result



```
import java.util.Scanner;
public class PowerOfNumber {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Take user input for the number
        System.out.print("Enter the number: ");
        int number = input.nextInt();

        // Take user input for the power
        System.out.print("Enter the power: ");
        int power = input.nextInt();

        // Initialize the result variable
        int result = 1;
        // Run a for loop from i = 1 to i <= power
        for (int i = 1; i <= power; i++) {
            // Multiply the result with the number and assign the
            value to result
            result *= number;
        }

        // Print the result
        System.out.println("The result of " + number + " raised to the
        power of " + power + " is " + result);

        // Close the scanner
        input.close();
    }
}
```

11. Create a program to find the factors of a number taken as user input.

**Hint =>**

- Get the input value for a variable named number.
- Run a **for** loop from  $i = 1$  to  $i < \text{number}$ .
- In each iteration of the loop, check if number is perfectly divisible by  $i$ .
- If true, print the value of  $i$ .

```
import java.util.Scanner;
public class Factors {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
        Scanner input = new Scanner(System.in);

        // Take user input for the number
        System.out.print("Enter an integer: ");
        int number = input.nextInt();
        System.out.println("The factors of " + number + " are:");

        // Run a for loop from i = 1 to i < number
        for (int i = 1; i < number; i++) {

            // Check if the number is perfectly divisible by i
            if (number % i == 0) {

                // Print the value of i
                System.out.println(i);
            }
        }
        // Close the scanner
        input.close();
    }
}
```

12. Create a program to find all the multiple of a number taken as user input below 100.

**Hint =>**

- Get input value for a variable named number.
- Run a **for** loop backward: from  $i = 100$  to  $i = 1$ .
- Inside the loop, check if  $i$  perfectly divide the number.
- If true, print the number and **continue** the loop.

```
import java.util.Scanner;
public class MultiplesBelowHundred {
    public static void main(String[] args) {

        // Create a Scanner object to take user input
```

```
Scanner input = new Scanner(System.in);

// Take user input for the number
System.out.print("Enter a number: ");
int number = input.nextInt();
System.out.println("Multiples of " + number + " below 100
are:");

// Run a for loop backward from i = 100 to i = 1
for (int i = 100; i >= 1; i--) {

    // Check if i perfectly divides the number
    if (i % number == 0) {

        // Print the value of i
        System.out.println(i);
    }
}
// Close the scanner
input.close();
}
```